Pranav Shukla

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Education

New York University Tandon School of Engineering **2024-Present**
M.S. Mechatronics and Robotics
Indian Institute of Technology Kharagpur **2017-2022**
B.Tech and M.Tech (Dual Degree) in Structural Engineering Specialization - Dept. of Civil Engineering
Minors: Computer Science and Engineering; Embedded Control, Software and Design

Skills

Languages Python, Bash, C++, VBA, SQL, APL, XML, JS
Development Git, CV2, React/Native, Selenium, Fenics, ROS
ML SAS, Tableau, Big Data, OpenAI API, Pytorch , Pydantic AI, VLA, Ollama, CrewAI, Lerobot, HPC, ACT, GROOT
Hardware Mechatronics, Control Systems, Prototyping
Platforms MATLAB, Solidworks, Gazebo, Abaqus, Staad, Verilog, Arduino, Raspian, MS Office

Projects/ Competitions

Emobodied AI Mobile Manipulator (Agentic Framework, VLA) . DEMO Video
- Developed a mobile manipulator with a lift, differential drive, a 5 dof robotic arm controlled by a vision-language model.
- The agent framework can create, schedule, and execute new tasks based on user text input. (Segmented VLA)
- A top-down navigation problem was approached through a gemini vision model with A* infused path planning and automated control through the VLM.
- Small Size Local VLMs, such as Moondream, were utilised to solve problems such as search and interact with a person/object of description.

Imitation Learning and Control with the SO-100 Robotic Arm . DEMO Video
- Developed an HPC pipeline to train, and load inference of a SO-100 robotic arm for imitation learning tasks.
- Used models such as ACT (Action Chunking Transformers), Diffusion, Groot, to imitate a variety of tasks with successful generalisation (after training on datasets recorded with teleoperation utilising the LeRobot framework).
- Tasks developed included: Handshake, Cleaning up a kitchen counter with a tablecloth, picking up cutlery and placing it in it's relevant bins, picking up a can and dropping it to the person described, and making a Sandwich.

RSSI based Indoor localisation system (ESP, microcontrollers)
- Designed a Wifi Enabled ESP-Based Localization for triangulation of ground bot position within a 10 cm error margin
- Used orientation specific isotropy, RSSI strength and experimentally validated distance mapping.

Inhomogeneity Detection using ANN for Inverse Cauchy type Problem (Numerical Methonds, ML)
- Designed an algorithm to detect damages inside structures using partial boundary data and domain equations
- Designed an ANN to solve a two dimensional time independent elliptic partial differential equation using the Dirichelet and Neumann boundary conditions on part of the boundary and with no information on the remaining boundary.
- Designed an alternating minimization based converging algorithm to detect changes in spatial material coefficients.

D.R.D.O. SASEs UAV Fleet Challenge (SLAM, gmaps, Networked Swarm)
- Developed an ad-hoc networked decentralised swarm of 4 drones for a autonomous vision based search and rescue task.
- Used Mavros and mesh routing protocol OLSR for adhoc communication and Google cloud APIs to create realtime maps.

Experience

Graduate Adjunct - Automatic Control Lab, New York University Tandon School of Engineering **2025-Present**
Matlab, Simulink, PID and LQR Control, Systems Analysis
- Teaching ME-UY 3411 covering experiments across Data acquisition, PID, LQR controls and real time system analysis

Business Analyst - Business Intelligence Unit, Axis Bank, Mumbai **2022-2024**
SAS, SQL, Data Analytics, Big Data, Version Control, Automata
- Developed pre-qualified banking product strategies using customer demographics, delinquency and bureau credit data; managed monthly releases, performance tracking dashboards, and automation pipeline.
- Ran simulations to identify database enhancements, then worked with Business, DE, DS, and Risk teams for actualization

Data Science Intern - Business Intelligence Unit, Axis Bank, Mumbai **2021- 2021**
Python, Tensorflow, Numpy, Big Data, Jupyter

- Implemented and validated 3 machine learning models (DT,MLP,Random Forest) for credit score prediction, optimized for biased datasets, ensuring accurate risk assessment and performance tracking using alternative data streams.

Embedded Team Head — Swarm Robotics Research Group, IIT Kharagpur **2019-2022**
ROS, Arduino, Linux, Control Systems, Computer Vision, Simulation, Mechatronics

- Deployed decentralised algorithms on 10 Kilobots by K-team such as, alloting ID, shape formation and segregation
- Designed prototypes for self assembling modular bots that can autonomously assemble/dissemble adding functionality
- Implemented ad-hoc networking for routerless P2P transmission and mesh routing protocols (BATMAN, OLSR) to enable automated rerouting in response to architecture changes like signal loss or node addition/removal.
- Led the Embedded team and oversaw the Deployment of electronic and mechanical stack for vision-based swarm-bots

Extracurricular

- **Teaching Assistant** for Courses: Monte Carlo Simulations and course: Construction, Planning and Management.
- Governor - Technology Literary Society, IIT KGP - Led a society of 80+ students, Editor of the campus magazine, Alankar
- Captain of Product Design Team IIT KGP GC, 2019-2020. In the NSS, ran rural clothes donation drives
- **Other Projects:** Slayer Exciter, Vision Based Gerontechnology, Panorama Image Stitching, VIO localisation